

$$\begin{aligned}
& -\frac{1}{4}(-g_1^2+g_2^2)+ \\
& -\frac{1}{16\pi^2}\left(-\frac{1}{4}g_1^4+\frac{1}{6}g_2^4+\frac{1}{12}\sum_p g_1^4\text{LF}_{2,0}[\mathbf{m}_p^D]-\frac{1}{2}g_1^2\overline{y}^{\text{PR}}y^{\text{PR}}\text{LF}_{2,0}[\mathbf{m}_p^D]+\frac{1}{4}\sum_p g_1^4\text{LF}_{2,0}[\mathbf{m}_p^P]-\right. \\
& -\frac{1}{2}g_1^2\overline{y}^{\text{PR}}y^{\text{PR}}\text{LF}_{2,0}[\mathbf{m}_e^r]+\frac{1}{8}(g_1^2+g_2^2)(2\overline{y}^{\text{PR}}y^{\text{PR}}+\sum_p(g_1^2-g_2^2))\text{LF}_{2,0}[\mathbf{m}_l^r]+ \\
& +\frac{1}{24}(-6\overline{y}^{\text{PR}}y^{\text{PR}}(g_1^2-3g_2^2)+6\overline{y}^{\text{PR}}y^{\text{PR}}(g_1^2+3g_2^2)+\sum_p(g_1^4-9g_2^4))\text{LF}_{2,0}[\mathbf{m}_q^P]+ \\
& +\frac{1}{2}\sum_p g_1^4\text{LF}_{2,0}[\mathbf{m}_p^P]-g_1^2\overline{y}^{\text{PR}}y^{\text{PR}}\text{LF}_{2,0}[\mathbf{m}_r^r]+\frac{1}{2}g_1^4\text{LF}_{2,2,-2}[\mathbf{m}_1,\tilde{\mu}]+ \\
& +\frac{1}{2}g_1^4m_1^2\text{LF}_{2,2,-1}[\mathbf{m}_1,\tilde{\mu}]+\frac{5}{2}g_2^4\text{LF}_{2,2,-2}[\mathbf{m}_2,\tilde{\mu}]+\frac{1}{2}g_2^4m_2^2\text{LF}_{2,2,-1}[\mathbf{m}_2,\tilde{\mu}]+ \\
& +\frac{2}{3}m_2\tilde{\mu}g_2^4(\overline{C_{\phi 2\phi 1}}+C_{\phi 2\phi 1})\text{LF}_{2,2,0}[\mathbf{m}_2,\tilde{\mu}]-\frac{1}{3}m_2\tilde{\mu}g_2^4(\overline{C_{\phi 2\phi 1}}+C_{\phi 2\phi 1})\text{LF}_{3,2,-1}[\mathbf{m}_2,\tilde{\mu}]+ \\
& +\frac{1}{2}g_1^2(-\overline{a}^{\text{PR}}a^{\text{PR}}+\tilde{\mu}^2\overline{y}^{\text{PR}}y^{\text{PR}})\text{LF}_{2,1,0}[\mathbf{m}_d^r,\mathbf{m}_q^P]+ \\
& +\frac{1}{2}g_1^2(-C_{\phi 2^2}\overline{a}^{\text{PR}}a^{\text{PR}}+C_{\phi 1^2}\tilde{\mu}^2\overline{y}^{\text{PR}}y^{\text{PR}})\text{LF}_{2,2,0}[\mathbf{m}_d^r,\mathbf{m}_q^P]- \\
& +3\overline{y}^{\text{PR}}y^{\text{SR}}\overline{y}^{\text{st}}y^{\text{pt}}\text{LF}_{1,1,0}[\mathbf{m}_d^r,\mathbf{m}_u^t]+\frac{1}{2}g_1^2(-\overline{a}_e^{\text{PR}}a_e^{\text{PR}}+\tilde{\mu}^2\overline{y}_e^{\text{PR}}y_e^{\text{PR}})\text{LF}_{2,1,0}[\mathbf{m}_e^r,\mathbf{m}_l^P]+ \\
& +\frac{1}{2}g_1^2(-C_{\phi 2^2}\overline{a}_e^{\text{PR}}a_e^{\text{PR}}+C_{\phi 1^2}\tilde{\mu}^2\overline{y}_e^{\text{PR}}y_e^{\text{PR}})\text{LF}_{2,2,0}[\mathbf{m}_e^r,\mathbf{m}_l^P]+ \\
& +\frac{1}{2}(g_2^2\overline{a}_e^{\text{PR}}a_e^{\text{PR}}-g_1^2\tilde{\mu}^2\overline{y}_e^{\text{PR}}y_e^{\text{PR}})\text{LF}_{2,1,0}[\mathbf{m}_l^P,\mathbf{m}_e^r]+\frac{1}{4}(g_1^2-g_2^2)(\overline{a}_e^{\text{PR}}a_e^{\text{PR}}+\tilde{\mu}^2\overline{y}_e^{\text{PR}}y_e^{\text{PR}}) \\
& \text{LF}_{3,1,-1}[\mathbf{m}_l^P,\mathbf{m}_e^r]+\frac{1}{2}(\tilde{\mu}\overline{y}_e^{\text{PR}}(C_{\phi 2\phi 1}a_e^{\text{PR}}(-g_1^2+g_2^2)-2C_{\phi 1^2}\tilde{\mu}g_2^2y_e^{\text{PR}})+ \\
& \overline{a}_e^{\text{PR}}(2C_{\phi 2^2}g_2^2a_e^{\text{PR}}+\overline{C_{\phi 2\phi 1}}\tilde{\mu}y_e^{\text{PR}}(-g_1^2+g_2^2)))\text{LF}_{3,1,0}[\mathbf{m}_l^P,\mathbf{m}_e^r]+ \\
& +\frac{1}{2}g_1^2(C_{\phi 2^2}\overline{a}_e^{\text{PR}}a_e^{\text{PR}}-C_{\phi 1^2}\tilde{\mu}^2\overline{y}_e^{\text{PR}}y_e^{\text{PR}})\text{LF}_{3,2,-1}[\mathbf{m}_l^P,\mathbf{m}_e^r]+ \\
& +\frac{1}{4}(\overline{a}_e^{\text{PR}}(3C_{\phi 2^2}a_e^{\text{PR}}(g_1^2-3g_2^2)+\overline{C_{\phi 2\phi 1}}\tilde{\mu}y_e^{\text{PR}}(6g_1^2-5g_2^2))+ \\
& \tilde{\mu}\overline{y}_e^{\text{PR}}(C_{\phi 2\phi 1}a_e^{\text{PR}}(6g_1^2-5g_2^2)+3C_{\phi 1^2}\tilde{\mu}y_e^{\text{PR}}(3g_1^2-g_2^2)))\text{LF}_{4,1,-1}[\mathbf{m}_l^P,\mathbf{m}_e^r]+ \\
& +\frac{1}{3}(\tilde{\mu}\overline{y}_e^{\text{PR}}(C_{\phi 2\phi 1}a_e^{\text{PR}}(-3g_1^2+2g_2^2)+3C_{\phi 1^2}\tilde{\mu}y_e^{\text{PR}}(-g_1^2+g_2^2))+ \\
& \overline{a}_e^{\text{PR}}(3C_{\phi 2^2}a_e^{\text{PR}}(-g_1^2+g_2^2)+\overline{C_{\phi 2\phi 1}}\tilde{\mu}y_e^{\text{PR}}(-3g_1^2+2g_2^2)))\text{LF}_{5,1,-2}[\mathbf{m}_l^P,\mathbf{m}_e^r]+ \\
& +\frac{1}{2}(\overline{a}_d^{\text{PR}}a_d^{\text{PR}}(-2g_1^2+3g_2^2)-g_1^2\tilde{\mu}^2\overline{y}_d^{\text{PR}}y_d^{\text{PR}})\text{LF}_{2,1,0}[\mathbf{m}_q^P,\mathbf{m}_d^r]+ \\
& +\frac{3}{4}(g_1^2-g_2^2)(\overline{a}_d^{\text{PR}}a_d^{\text{PR}}+\tilde{\mu}^2\overline{y}_d^{\text{PR}}y_d^{\text{PR}})\text{LF}_{3,1,-1}[\mathbf{m}_q^P,\mathbf{m}_d^r]+ \\
& +\frac{1}{2}(\tilde{\mu}\overline{y}_d^{\text{PR}}(3C_{\phi 2\phi 1}a_d^{\text{PR}}(-g_1^2+g_2^2)-2C_{\phi 1^2}\tilde{\mu}g_1^2y_d^{\text{PR}})+ \\
& \overline{a}_d^{\text{PR}}(2C_{\phi 2^2}a_d^{\text{PR}}(-2g_1^2+3g_2^2)+3\overline{C_{\phi 2\phi 1}}\tilde{\mu}y_d^{\text{PR}}(-g_1^2+g_2^2)))\text{LF}_{3,1,0}[\mathbf{m}_q^P,\mathbf{m}_d^r]+ \\
& +\frac{1}{2}g_1^2(C_{\phi 2^2}\overline{a}_d^{\text{PR}}a_d^{\text{PR}}-C_{\phi 1^2}\tilde{\mu}^2\overline{y}_d^{\text{PR}}y_d^{\text{PR}})\text{LF}_{3,2,-1}[\mathbf{m}_q^P,\mathbf{m}_d^r]+ \\
& +\frac{3}{4}(\overline{a}_d^{\text{PR}}(C_{\phi 2^2}a_d^{\text{PR}}(7g_1^2-9g_2^2)+\overline{C_{\phi 2\phi 1}}\tilde{\mu}y_d^{\text{PR}}(6g_1^2-5g_2^2))+ \\
& \tilde{\mu}\overline{y}_d^{\text{PR}}(C_{\phi 2\phi 1}a_d^{\text{PR}}(6g_1^2-5g_2^2)+C_{\phi 1^2}\tilde{\mu}y_d^{\text{PR}}(5g_1^2-3g_2^2)))\text{LF}_{4,1,-1}[\mathbf{m}_q^P,\mathbf{m}_d^r]+ \\
& +(\tilde{\mu}\overline{y}_d^{\text{PR}}(C_{\phi 2\phi 1}a_d^{\text{PR}}(-3g_1^2+2g_2^2)+3C_{\phi 1^2}\tilde{\mu}y_d^{\text{PR}}(-g_1^2+g_2^2))+ \\
& \overline{a}_d^{\text{PR}}(3C_{\phi 2^2}a_d^{\text{PR}}(-g_1^2+g_2^2)+\overline{C_{\phi 2\phi 1}}\tilde{\mu}y_d^{\text{PR}}(-3g_1^2+2g_2^2)))\text{LF}_{5,1,-2}[\mathbf{m}_q^P,\mathbf{m}_d^r]- \\
& +3\overline{y}_d^{\text{PR}}y_d^{\text{SR}}\overline{y}_d^{\text{st}}y_d^{\text{pt}}\text{LF}_{1,1,0}[\mathbf{m}_q^P,\mathbf{m}_l^S]+\frac{1}{4}(g_1^2+3g_2^2)(\overline{a}_u^{\text{PR}}a_u^{\text{PR}}-\tilde{\mu}^2\overline{y}_u^{\text{PR}}y_u^{\text{PR}}) \\
& \text{LF}_{2,1,0}[\mathbf{m}_q^P,\mathbf{m}_u^r]+\frac{1}{4}(\overline{a}_u^{\text{PR}}(C_{\phi 1^2}a_u^{\text{PR}}(g_1^2+3g_2^2)+2\overline{C_{\phi 2\phi 1}}\tilde{\mu}g_2^2y_u^{\text{PR}})+ \\
& \tilde{\mu}\overline{y}_u^{\text{PR}}(2C_{\phi 2\phi 1}g_2^2a_u^{\text{PR}}-C_{\phi 2^2}\tilde{\mu}y_u^{\text{PR}}(g_1^2+3g_2^2)))\text{LF}_{2,2,0}[\mathbf{m}_q^P,\mathbf{m}_u^r]- \\
& +\frac{1}{4}\tilde{\mu}g_2^2(C_{\phi 2\phi 1}\overline{y}_u^{\text{PR}}a_u^{\text{PR}}+\overline{C_{\phi 2\phi 1}}y_u^{\text{PR}}\overline{a}_u^{\text{PR}})\text{LF}_{3,2,-1}[\mathbf{m}_q^P,\mathbf{m}_u^r]+ \\
& +\frac{1}{4}(\overline{a}_u^{\text{PR}}a_u^{\text{PR}}(-7g_1^2+3g_2^2)+\tilde{\mu}^2\overline{y}_u^{\text{PR}}y_u^{\text{PR}}(g_1^2+3g_2^2))\text{LF}_{2,1,0}[\mathbf{m}_u^r,\mathbf{m}_q^P]+ \\
& +\frac{3}{4}(g_1^2-g_2^2)(\overline{a}_u^{\text{PR}}a_u^{\text{PR}}+\tilde{\mu}^2\overline{y}_u^{\text{PR}}y_u^{\text{PR}})\text{LF}_{3,1,-1}[\mathbf{m}_u^r,\mathbf{m}_q^P]+ \\
& +\frac{1}{2}(\overline{a}_u^{\text{PR}}(C_{\phi 1^2}a_u^{\text{PR}}(-7g_1^2+3g_2^2)+\overline{C_{\phi 2\phi 1}}\tilde{\mu}y_u^{\text{PR}}(-3g_1^2$$